

Machine Learning-Based Detection of Phishing URLs Using Lexical URL Features

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Abstract— The phishing web pages are still a substantial threat, as they imitate real websites and try to steal your user credentials and other sensitive files such as financial information. In this project, we investigated the sensitivity of lightweight URL-based lexical features and their use with six engineered ratio and binary indicators for efficient phishing detection and their correlation to classical supervised machine learning approaches. The experiments utilize the Kaggle Web Page Phishing Dataset comprising 100,077 rows, 19 URL-based numerical features, and one binary target label. By removing 78,186 duplicate rows, we have 21,891 unique records that contribute to the model workflow. A total of six additional columns are derived to model symbol density, redirection presence, and URL depth. The pipeline involves data checking, duplicate handling, feature construction, stratified train/test splitting, feature scaling where applicable, five-fold cross-validation, hyperparameter tuning on each of the models with GridSearchCV, and finally, held-out test evaluation. There are four evaluated models: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The best test accuracy was obtained by Gradient Boosting with an accuracy of 0.8198, precision 0.8505, recall 0.9118, F1-score 0.8801, and an ROC-AUC of 0.8873. All these models had F1-scores between 0.853 and 0.880, which showed that embedding feature engineering and strict hyperparameter tuning was able to provide stable and consistent and competitive performance across the model conditions examined.

I. INTRODUCTION

Phishing threats are a big problem in cybersecurity because they send consumers to false web pages, do website page fraud, which drives internet user click into the bad page. On the other hand, the only issue is not that there is malicious content that users have visited. Attackers are usually used to bait with multiple bait types such as URLs, domain signature, redirect behavior to mislead a victim. Since phishing attacks are dynamic and scale, automated detection is a vital measure of defense. URL-based detection is attractive due to the fact that it is lightweight and it does not have to download content from a webpage, run scripts or query a more sophisticated external

source such as WHOIS or DNS. Some of these lexical properties, such as URL's length, number of dots, number of hyphens, etc., might help to discern abnormal behavior. Furthermore, these characteristics are easy to extract and analyze using batch machine learning workflow. In this project, we propose phishing detection as supervised binary classification. We label the URL as either a legitimate or a phishing URL and use data-rich attributes to complete both training and evaluation of that URL, and measure the performance with accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis. Recall is especially critical here: a missed phishing page is a deadlier threat than a false alarm; so recall is absolutely critical. The main contributions of this work are as follows:

1. A reproducible Python implementation for phishing URL detection based on lexical URL features.
2. Comparing four classical and ensemble machine learning methods via a common evaluation flow.
3. Class balance, feature relationships across a number of class representations, confusion matrix behavior, as well as recall/F1 trade-offs may be examined in a structured fashion.
4. Analyses of related literature for phishing URL detection with similar lexical features.

II. RELATED WORK

II. RELATED WORK. Blum et al. examined phishing URL detection via lexical features in the field of online learning techniques [2]. By extracting the phishing signals directly from the URL text and not including the whole of the page content, they found that useful phishing signals can be derived from the URL text as well. Although related in its focus on lexical features, this research differs in dataset choice and overall comparison of batch classifiers. Sahingoz et al. explored data from the lexical and host-based signals from URLs to perform machine learning based phishing detection [3]. In their reported experiments, Random Forest with NLP-based features on a large URL dataset (36,400 legitimate and 37,175 phishing URLs)

achieved 97.98% accuracy. This is a more focused project; the scope of the project is limited to the Kaggle Web Page Phishing Dataset, and it also involves an easy reproducible pipeline using URL lexical features. Awasthi and Goel compared base and ensemble classifiers on two URL-feature phishing datasets that have 30 structural URL features and 10-fold cross-validation. Their base classifiers (i.e., Logistic Regression, Decision Tree, and SVM) attained an accuracy rate of 93% to 96%, which is methodologically significant because they use similar classifier families and URL-related features, but the datasets and feature sets differ. Alnemari and Alshammari [5] evaluated Random Forest, Decision Tree, SVM, and ANN on 11,055 UCI phishing website samples based on 30 URL-structural features. Random Forest fared the best with 97.3% accuracy while SVM and Decision Tree were 94–96%. Note that their article is helpful as it has the same classes of classical model families as the present work in a similar URL-feature dataset. Although this work utilises Kaggle Web Page Phishing Dataset, the two studies compared are based upon two different phishing URL datasets. The comparisons hence focus solely on methodology, feature types, type of classifier, and reported metric ranges as opposed to using the same set data to benchmark directly.

III. EXPERIMENTAL SETUP

A. Dataset Description

We use the Kaggle Web Page Phishing Dataset [1] for this project. In the local CSV, the dataset is composed of 100,077 rows and 20 columns before cleaning. The dataset has nineteen data columns, which are numerical features of URLs, and the target column is phishing (1 being phishing and 0 being legitimate). After removing the duplicates for modeling, there are 21,891 unique rows. Following that, the original columns are then amended with six features, for a total of 25 features. Table I lists the input features and the pipeline's target variable.

TABLE I
FEATURES USED IN THE PHISHING DETECTION PIPELINE

Feature	Type	Description
url_length	Numerical	Total characters in the URL
n_dots	Numerical	Count of . characters
n_hyphens	Numerical	Count of hyphen characters
n_underline	Numerical	Count of _ characters
n_slash	Numerical	Count of / characters
n_questionmark	Numerical	Count of ? characters
n_equal	Numerical	Count of = characters
n_at	Numerical	Count of @ characters
n_and	Numerical	Count of & characters
n_exclamation	Numerical	Count of ! characters
n_space	Numerical	Count of space characters
n_tilde	Numerical	Count of ~ characters
n_comma	Numerical	Count of , characters
n_plus	Numerical	Count of + characters
n_asterisk	Numerical	Count of * characters
n_hashtag	Numerical	Count of # characters
n_dollar	Numerical	Count of \$ characters
n_percent	Numerical	Count of % characters
n_redirection	Numerical	Count of redirect-like markers
total_special_chars	Engineered	Sum of all special character counts
symbol_ratio	Engineered	total_special_chars / (url_length+1)
has_redirect	Engineered	Binary: 1 if n_redirection > 0
has_at	Engineered	Binary: 1 if n_at > 0
slash_dot_ratio	Engineered	n_slash / (n_dots+1)
url_length_log	Engineered	log(url_length + 1)
phishing	Binary	1 = phishing, 0 = legitimate (target)

No missing data existed in this cleaned dataset. The removal of duplicates resulted in a class representation with 6,019 legitimate URLs and 15,872 phishing URLs, which amounts to about 27.5% legitimate and 72.5% phishing URLs.



Fig. 1. Class distribution after deduplication.

B. Exploratory Data Analysis

This exploratory analysis considers class balance, feature distributions, and feature-target associations. From the correlation analysis, we observed that `n_slash`, `url_length`, `n_equal`, `n_redirection`, and `n_dots` have the strongest absolute linear relationships with the phishing label in the dataset. In particular, there are some count-based features with long tails, including `url_length`, `n_hypens`, and `n_slash`, which are somewhat skewed.

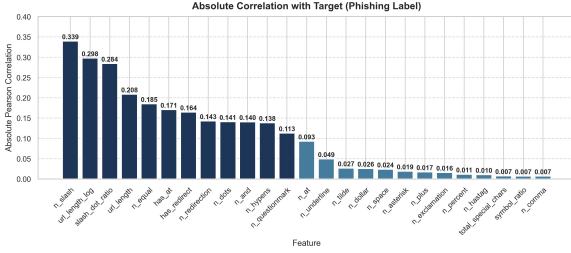


Fig. 2. Absolute correlation of features with the phishing label.

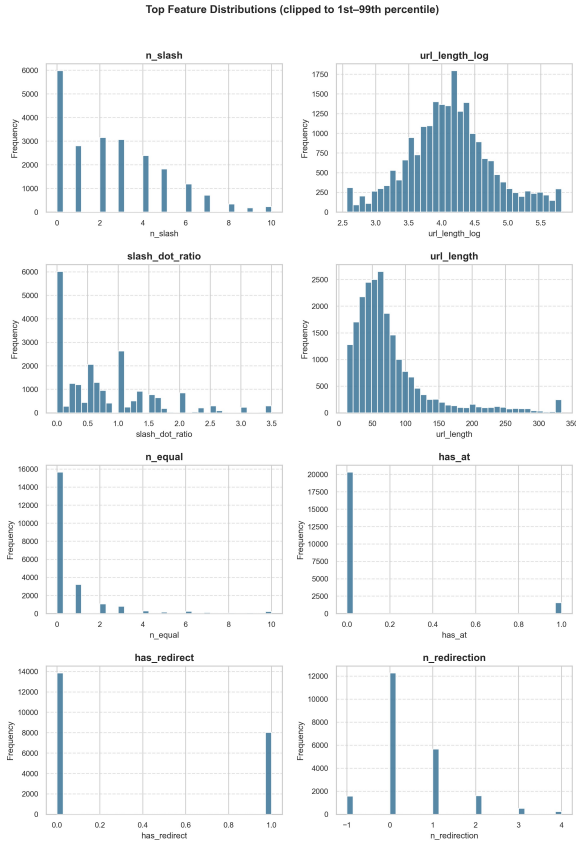


Fig. 3. Top Feature distributions.

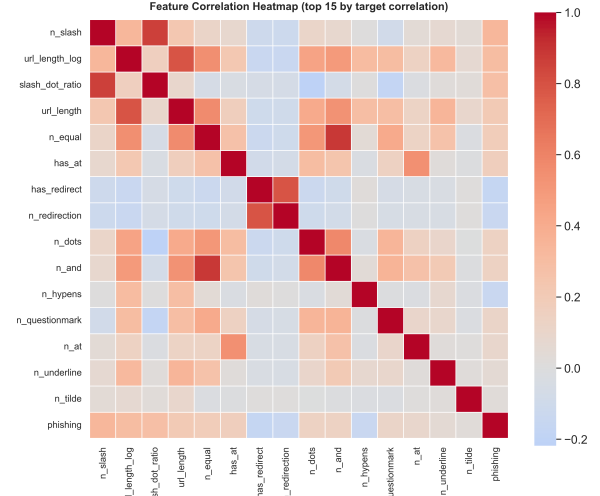


Fig. 4. Feature correlation heatmap.

C. Data Preprocessing

Duplicate rows are removed once before separation of the model, which is a lot important in this context where more than three quarters of the original rows are duplicates. Six features are added following deduplication: a total special character count; a symbol-to-length ratio; binary redirect and at-symbol flags; a slash-to-dot ratio; and a log-transformed URL length. In this paper, we have applied a stratified 80/20 train/test split to keep class proportions. Feature scaling only applies to Logistic Regression.

D. Evaluation Metrics

Accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis are used for the project. Recall is particularly important as false negatives correspond to phishing pages that evade detection and thus have the greatest security risk.

IV. MACHINE LEARNING ALGORITHMS

The project evaluate four classifiers for the spectrum from simple linear models to complex boosting ensembles. Each model is tuned independently with GridSearchCV using 5-fold stratified cross-validation scored on F1.

A. Logistic Regression

Logistic Regression is included as the linear baseline. It models the log-odds of the phishing class as a weighted sum of the input features. Standard scaling is applied upstream in the pipeline. The regularization parameter C is tuned across 0.01, 0.1, 1, and 10.

B. Decision Tree

The Decision Tree classifier recursively partitions the feature space by selecting the split that maximizes information gain at each node. Maximum depth, minimum samples per split, and minimum samples per leaf are all tuned to control overfitting.

C. Random Forest

Random Forest is a bagging ensemble that trains a collection of decision trees on bootstrapped subsets of the training data with randomized feature subsets at each split. The number of estimators, maximum depth, and leaf size parameters are tuned via grid search.

D. Gradient Boosting

Gradient Boosting is a sequential boosting ensemble that builds trees one at a time, each correcting the residual errors of the previous ensemble. Optimal learning rate, number of estimators, maximum tree depth, and subsampling ratio are adjusted together to balance the bias and variance parameters.

V. RESULTS AND ANALYSIS

A. Model Performance

Table II summarizes the held-out test results for all four models after hyperparameter tuning using GridSearchCV.

TABLE II

TEST PERFORMANCE AFTER HYPERPARAMETER TUNING

Model	Accuracy	Precision	Recall	F1	AUC
Gradient Boosting	0.8198	0.8505	0.9118	0.8801	0.8873
Random Forest	0.8102	0.8429	0.9074	0.8740	0.8793
Decision Tree	0.8022	0.8476	0.8866	0.8667	0.8478
Logistic Regression	0.7753	0.8101	0.9014	0.8533	0.8329

Gradient Boosting was found to be the best overall, with the best accuracy, F1-score, and ROC-AUC. Random Forest came in a close second with F1 = 0.874. Decision Tree significantly increased from an untuned F1 of 0.791 to 0.867 after hyperparameter tuning. The overall F1 range between 0.853 to 0.880 shows stable and competitive results across the evaluated model set. Since 72.5% of the deduplicated dataset are phishing samples, accuracy alone doesn't cut it; hence, F1-score, recall, ROC-AUC, and confusion matrix behavior are emphasized during the whole evaluation.

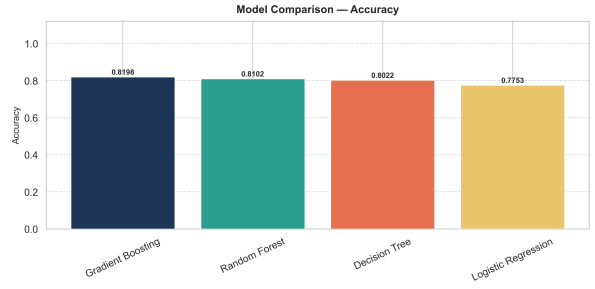


Fig. 5. Accuracy comparison across models.

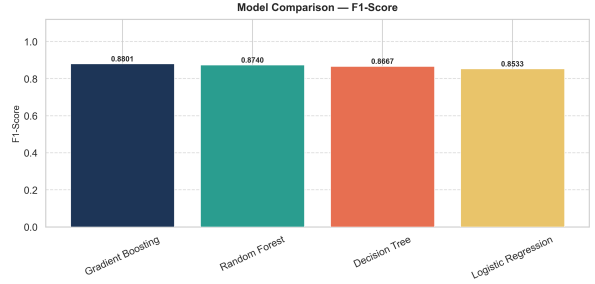


Fig. 6. F1-score comparison across models.

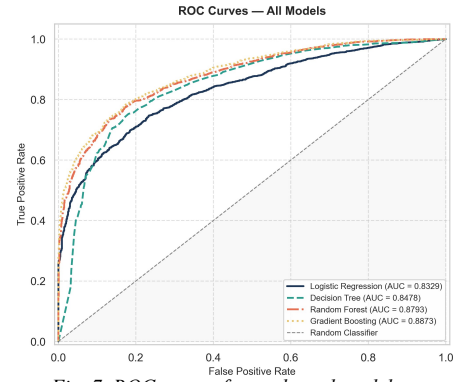


Fig. 7. ROC curves for evaluated models.

B. Best Model Confusion Matrix

Table III presents the confusion matrix for Gradient Boosting on the held-out test set.

TABLE III

CONFUSION MATRIX OF THE GRADIENT BOOSTING MODEL

Actual / Predicted	Legitimate	Phishing
Legitimate	695	509
Phishing	280	2895

The model correctly identifies 2,895 of 3,175 phishing URLs, yielding recall = 0.912. The 509 false positives represent legitimate URLs flagged as phishing, a security-oriented trade-off, although the false-positive rate should be reduced before real-world deployment.

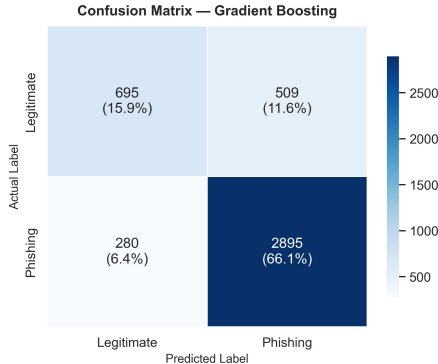


Fig. 8. Confusion matrix.

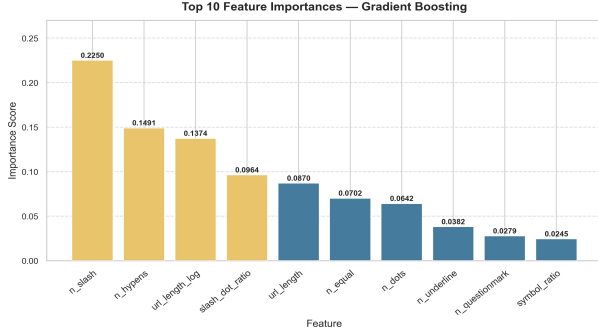


Fig. 9. Top feature importances for Gradient Boosting.

C. Discussion

1. Gradient Boosting achieved $F1 = 0.8801$ and $ROC-AUC = 0.8873$, the best results overall.
2. The decision tree improved substantially ($F1$: 0.791 to 0.867), demonstrating the importance of hyperparameter tuning.
3. All four models are competitive with $F1$ spanning 0.853-0.880.
4. Engineered features `symbol_ratio` and `slash_dot_ratio` appear in the top feature importances of tree-based models (Fig. 9).
5. Results differ from [4] and [5] in absolute accuracy due to stricter deduplication and a more limited feature set, but the evaluated model families are methodologically comparable.

VI. DISCUSSION AND LIMITATIONS

This project employs URL lexical features augmented with six engineered indicators, which make it a lightweight method and fast. The engineered features, particularly the symbol density ratio, slash-to-dot ratio, and binary redirect and at-symbol flags, add additional discriminative signals that supplement the raw counts and leverage linear models that cannot independently learn such relationships.

Even with feature engineering, lexical features alone may overlook sophisticated phishing pages that employ cleaner-looking URLs or rely on infrastructure-level deception. The compared studies that report higher accuracy generally use richer feature sets that include structural, host-based, or content-based attributes. The current pipeline intentionally restricts itself to URL-text features to ensure reproducibility and computational simplicity.

The freshness of the datasets is another constraint. A model trained on a static public dataset might not represent the newest phishing tactics. The current pipeline does not scan webpage content, WHOIS data, DNS signals, SSL info, or dynamic behavior. Real-world deployment would need external validation on more recent data. One other notable limitation of the provided CSV is that the structure is duplicate as well. Removing duplicates decreased the working dataset from 100,077 rows to 21,891 unique rows. Results from experiments that do not deduplicate are likely to appear artificially stronger.

VII. CONCLUSION

This work is a complete machine learning model for phishing URL detection using a public dataset of lexical URL attributes augmented with six engineered indicators. Experiments were executed after deduplication on 21,891 unique URL records. All four models were tuned through GridSearchCV, and hyperparameter optimization achieved a significant improvement in Decision Tree performance ($F1$: 0.791 to 0.867) and modest but also reasonably consistent gains throughout the ensemble models. Gradient Boosting was among the models evaluated, achieving 0.8198 accuracy, the best overall test performance, with 0.8505 precision, 0.9118 recall, 0.8801 $F1$ -score, and 0.8873 $ROC-AUC$. These results are methodologically consistent with previous studies but with lower absolute accuracy, probably due to strict duplication removal and the use of a small lexical feature set in this project. URL lexicon together with derived indicators may offer effective anti-phishing detection without content-based inspection. The approach should be validated in future work on newer datasets and extend the set of features with host, DNS, SSL, or content-based signals.

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